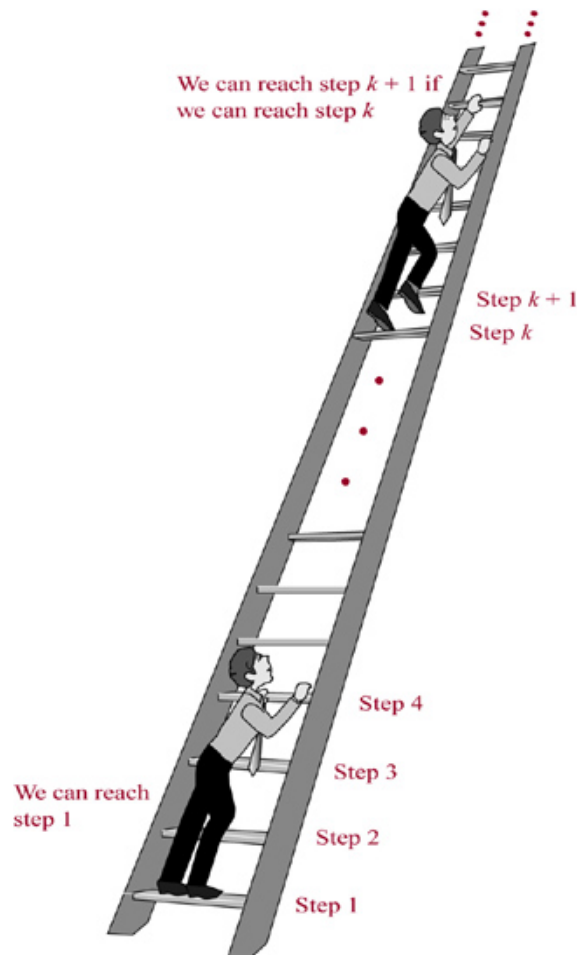


Discrete Mathematics

5.1 Mathematical induction

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- Want to know whether we can reach *every* step of this ladder
 - We can reach *first* rung of the ladder
 - If we can reach *a particular* rung of the ladder, then we can reach *the next* rung
- **Mathematical induction:** show that $p(n)$ is true for **every** positive integer n

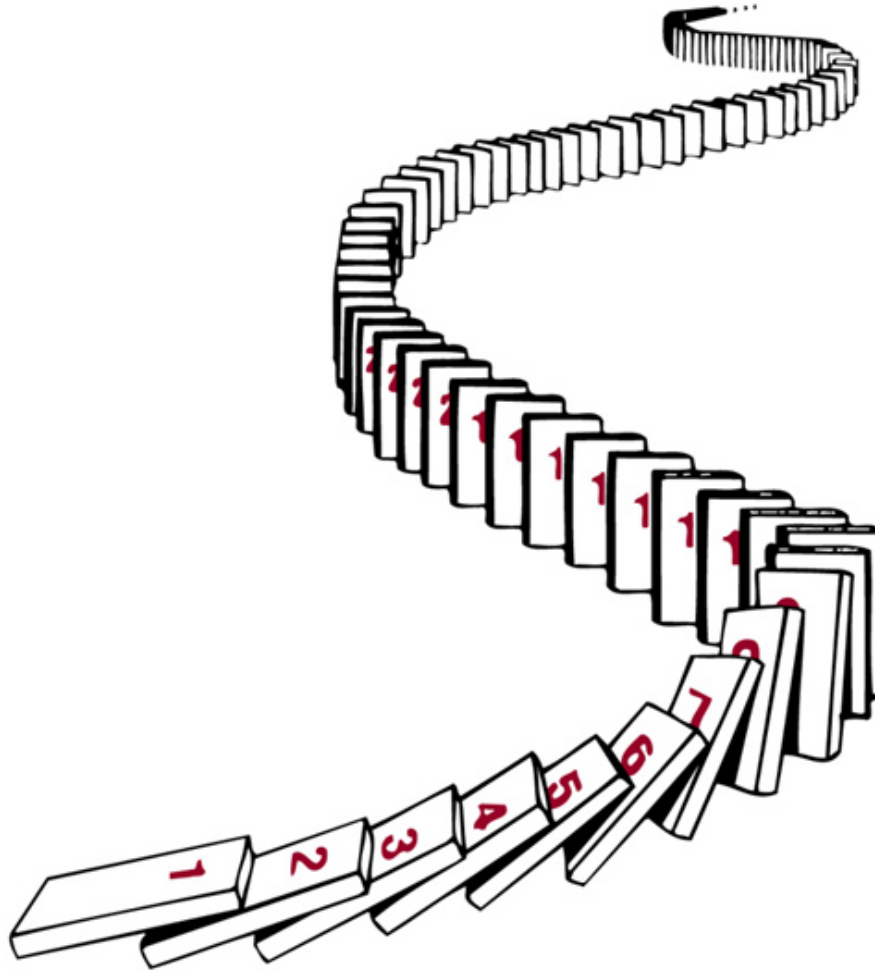
Mathematical induction

- Two steps
 - **Basis step**: show that $p(1)$ is true
 - **Inductive step**: show that for all positive integers k , if $p(k)$ is true, then $p(k+1)$ is true. That is, we show $p(k) \rightarrow p(k+1)$ for all positive integers k
- The assumption $p(k)$ is true is called the **inductive hypothesis**
- Proof technique:
$$[p(1) \wedge \forall k(p(k) \rightarrow p(k+1))] \rightarrow \forall n p(n)$$

Analogy

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Example

- Show that $1+2+\dots+n=n(n+1)/2$, if n is a positive integer
 - Let $p(n)$ be the proposition that $1+2+\dots+n=n(n+1)/2$
 - Basis step: $p(1)$ is true, because $1=1*(1+1)/2$
 - Inductive step: Assume $p(k)$ is true for an arbitrary k . That is, $1+2+\dots+k=k(k+1)/2$

We must show that $1+2+\dots+(k+1)=(k+1)(k+2)/2$

From $p(k)$, $1+2+\dots+k+(k+1)=k(k+1)/2+(k+1)=(k+1)(k+2)/2$

which means $p(k+1)$ is true
 - We have completed the basic and inductive steps, so by mathematical induction we know that $p(n)$ is true for all positive integers n . That is $1+2+\dots+n=n(n+1)/2$

Example

- Conjecture a formula for the sum of the first n positive odd integers. Then prove the conjecture using mathematical induction
- $1=1$, $1+3=4$, $1+3+5=9$, $1+3+5+7=16$,
 $1+3+5+7+9=25$
- It is reasonable to conjecture the sum of first n odd integers is n^2 , that is, $1+3+5+\dots+(2n-1)=n^2$
- We need a method to prove whether this conjecture is correct or not

Example

- Let $p(n)$ denote the proposition
- Basic step: $p(1)=1^2=1$
- Inductive steps: Assume that $p(k)$ is true, i.e.,
 $1+3+5+\dots+(2k-1)=k^2$

We must show $1+3+5+\dots+(2k+1)=(k+1)^2$ is true for $p(k+1)$

Thus, $1+3+5+\dots+(2k-1)+(2k+1)=k^2+2k+1=(k+1)^2$ which means $p(k+1)$ is true

(Note $p(k+1)$ means $1+3+5+\dots+(2k+1)=(k+1)^2$)

- We have completed both the basis and inductive steps. That is, we have shown $p(1)$ is true and $p(k) \rightarrow p(k+1)$
- Consequently, $p(n)$ is true for all positive integers n

Example

- Use mathematical induction to show that $1+2+2^2+\dots+2^n=2^{n+1}-1$
- Let $p(n)$ be the proposition: $1+2+2^2+\dots+2^n=2^{n+1}-1$
- Basis step: $p(0)=2^{0+1}-1=1$
- Inductive step: Assume $p(k)$ is true, i.e., $1+2+2^2+\dots+2^k=2^{k+1}-1$

It follows

$(1+2+2^2+\dots+2^k)+2^{k+1}=(2^{k+1}-1)+2^{k+1}=2*2^{k+1}-1=2^{k+2}-1$ which means $p(k+1)$: $1+2+2^2+\dots+2^{k+1}=2^{k+2}-1$ is true

- We have completed both the basis and inductive steps. By induction, we show that $1+2+2^2+\dots+2^n=2^{n+1}-1$

Example

- In the previous step, $p(0)$ is the basis step as the theorem is true $\forall n \geq 0$ $p(n)$ for all non-negative integers
- To use mathematical induction to show that $p(n)$ is true for $n=b, b+1, b+2, \dots$ where b is an integer other than 1, we show that $p(b)$ is true, and then $p(k) \rightarrow p(k+1)$ for $k=b, b+1, b+2, \dots$
- Note that b can be negative, zero, or positive

Example

- Use induction to show $\sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r-1}$ if $r \neq 1$
- Basis step: $p(0)$ is true as $\frac{ar^1 - a}{r-1} = a$
- Inductive step: assume $\sum_{j=0}^k ar^j = \frac{ar^{k+1} - a}{r-1}$ if $r \neq 1$

$$\begin{aligned}
 \sum_{j=0}^{k+1} ar^j &= a + ar + \dots + ar^k + ar^{k+1} \\
 &= \frac{ar^{k+1} - a}{r-1} + ar^{k+1} \\
 &= \frac{ar^{k+1} - a + ar^{k+2} - ar^{k+1}}{r-1} \\
 &= \frac{ar^{k+2} - a}{r-1}
 \end{aligned}$$

- So $p(k+1)$ is true. By induction, $p(n)$ is true for all nonnegative integers

Example

- Use induction to show that $n < 2^n$ for $n > 0$
- Basis step: $p(1)$ is true as $1 < 2^1 = 2$
- Inductive step: Assume $p(k)$ is true, i.e., $k < 2^k$

We need to show $k+1 < 2^{k+1}$

$$k+1 < 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$$

Thus $p(k+1)$ is true

- We complete both basis and inductive steps, and show that $p(n)$ is true for all positive integers n

Example

- Use induction to show that $2^n < n!$ for $n \geq 4$
- Let $p(n)$ be the proposition, $2^n < n!$ for $n \geq 4$
- Basis step: $p(4)$ is true as $2^4 = 16 < 4! = 24$
- Inductive step: Assume $p(k)$ is true, i.e., $2^k < k!$ for $k \geq 4$. We need to show that $2^{k+1} < (k+1)!$ for $k \geq 4$
$$2^{k+1} = 2 * 2^k < 2 * k! < (k+1) * k! = (k+1)!$$

This shows $p(k+1)$ is true when $p(k)$ is true
- We have completed basis and inductive steps.
By induction, we show that $p(n)$ is true for $n \geq 4$

Example

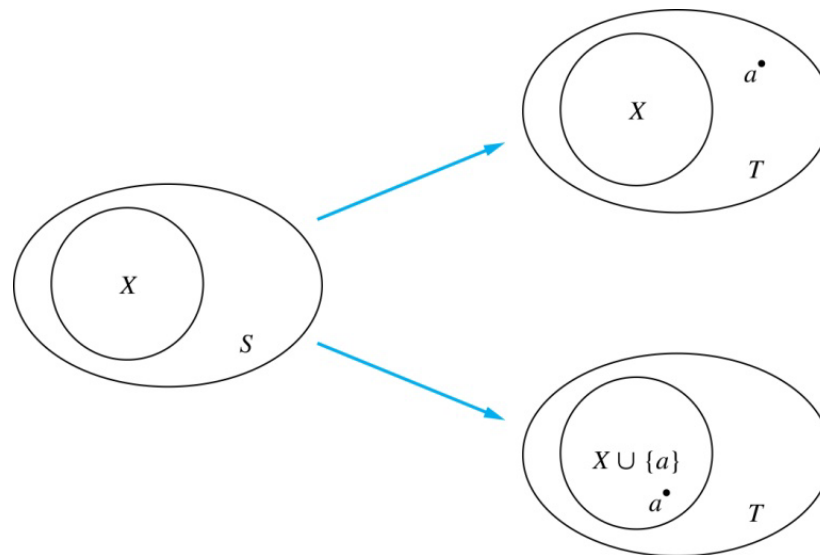
- Show that $n^3 - n$ is divisible by 3 when n is positive
- Basis step: $p(1)$ is true as $1 - 1 = 0$ is divisible by 3
- Inductive step: Suppose $p(k) = k^3 - k$ is true, we must show that $(k+1)^3 - (k+1)$ is divisible by 3
$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - (k+1) = (k^3 - k) + 3(k^2 + k)$$
As both terms are divisible by 3, $(k+1)^3 - (k+1)$ is divisible by 3
- We have completed both the basis and inductive steps. By induction, we show that $n^3 - n$ is divisible by 3 when n is positive

Example

- Show that if S is a finite set with n elements, then S has 2^n subsets
- Let $p(n)$ be the proposition that a set with n elements has 2^n subsets
- Basis step: $p(0)$ is true as a set with zero elements, the empty set, has exactly 1 subset
- Inductive step: Assume $p(k)$ is true, i.e., S has 2^k subsets if $|S|=k$.

Example

- Let T be a set with $k+1$ elements. So, $T=S\cup\{a\}$, and $|S|=k$
- For each subset X of S , there are exactly two subsets of T , i.e., X and $X \cup \{a\}$
- Because there are 2^k subsets of S , there are $2 \cdot 2^k = 2^{k+1}$ subsets of T . This finishes the inductive step



Example

- Use mathematical induction to show one of the De Morgan's law: $\overline{\bigcap_{j=1}^n A_j} = \bigcup_{j=1}^n \overline{A_j}$ where $A_1, A_2, \dots, A_n$ are subsets of a universal set U , and $n \geq 2$

$$\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$$

- Basis step: (proved Section 2.2, page 131)

$$\overline{\bigcap_{j=1}^k A_j} = \bigcup_{j=1}^k \overline{A_j}$$

- Inductive step: Assume $\overline{\bigcap_{j=1}^k A_j} = \bigcup_{j=1}^k \overline{A_j}$ is true for $k \geq 2$

$$\begin{aligned} \overline{\bigcap_{j=1}^{k+1} A_j} &= \overline{\left(\bigcap_{j=1}^k A_j \right) \cap A_{k+1}} = \overline{\left(\bigcap_{j=1}^k A_j \right)} \cup \overline{A_{k+1}} \\ &= \left(\bigcup_{j=1}^k \overline{A_j} \right) \cup \overline{A_{k+1}} = \bigcup_{j=1}^{k+1} \overline{A_j} \end{aligned}$$

Axioms for the set of positive integers

- See appendix 1
- Axiom 1: The number 1 is a positive integer
- Axiom 2: If n is a positive integer, then $n+1$, the successor of n , is also a positive integer
- Axiom 3: Every positive integer other than 1 is the successor of a positive integer
- Axiom 4: **Well-ordering property** Every non-empty subset of the set of positive

Why mathematical induction is valid?

- From mathematical induction, we know $p(1)$ is true and the proposition $p(k) \rightarrow p(k+1)$ is true for all positive integers
- To show that $p(n)$ must be true for all positive integers, assume that there is at least one positive integer such that $p(n)$ is false
- Then the set S of positive integers for which $p(n)$ is false is non-empty
- By well-ordering property, S has a least element, which is denoted by m
- We know that m cannot be 1 as $p(1)$ is true
- Because m is positive and greater than 1, $m-1$ is a positive integer

Why mathematical induction is valid?

- Because $m-1$ is less than m , it is not in S
- So $p(m-1)$ must be true
- As the conditional statement $p(m-1) \rightarrow p(m)$ is also true, it must be the case that $p(m)$ is true
- This contradicts the choice of m
- Thus, $p(n)$ must be true for every positive integer n

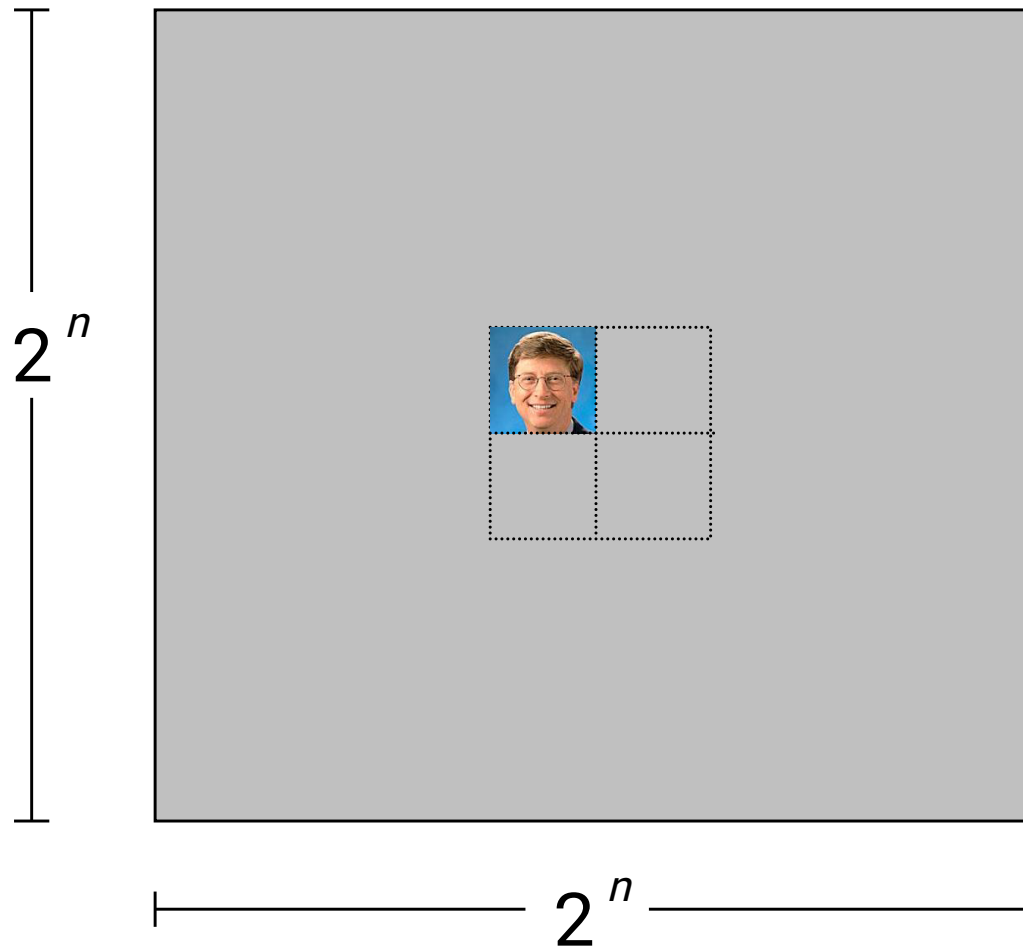
Template for inductive proof

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form “for all $n \geq b$, $P(n)$ ” for a fixed integer b .
2. Write out the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words “Inductive Step.”
4. State, and clearly identify, the inductive hypothesis, in the form “assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
5. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
6. Prove the statement $P(k + 1)$ making use the assumption $P(k)$. Be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.
7. Clearly identify the conclusion of the inductive step, such as by saying “this completes the inductive step.”
8. After completing the basis step and the inductive step, state the conclusion, namely that by mathematical induction, $P(n)$ is true for all integers n with $n \geq b$.

Puzzle

Goal: tile the squares, except one in the middle for Bill.

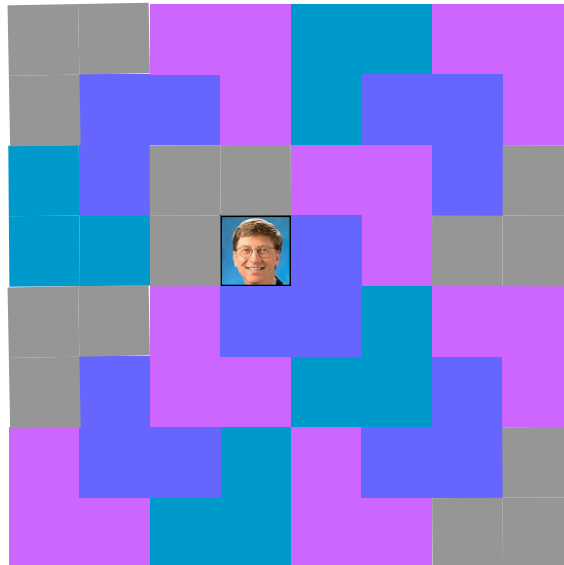


Puzzle

There are only L-shaped tiles covering three squares:



For example, for 8 x 8 puzzle might tile for Bill this way:



Puzzle

Theorem: For any $2^n \times 2^n$ puzzle, there is a tiling with Bill in the middle.

Proof: (by induction on n)

$P(n) ::=$ can tile $2^n \times 2^n$ with Bill in middle.

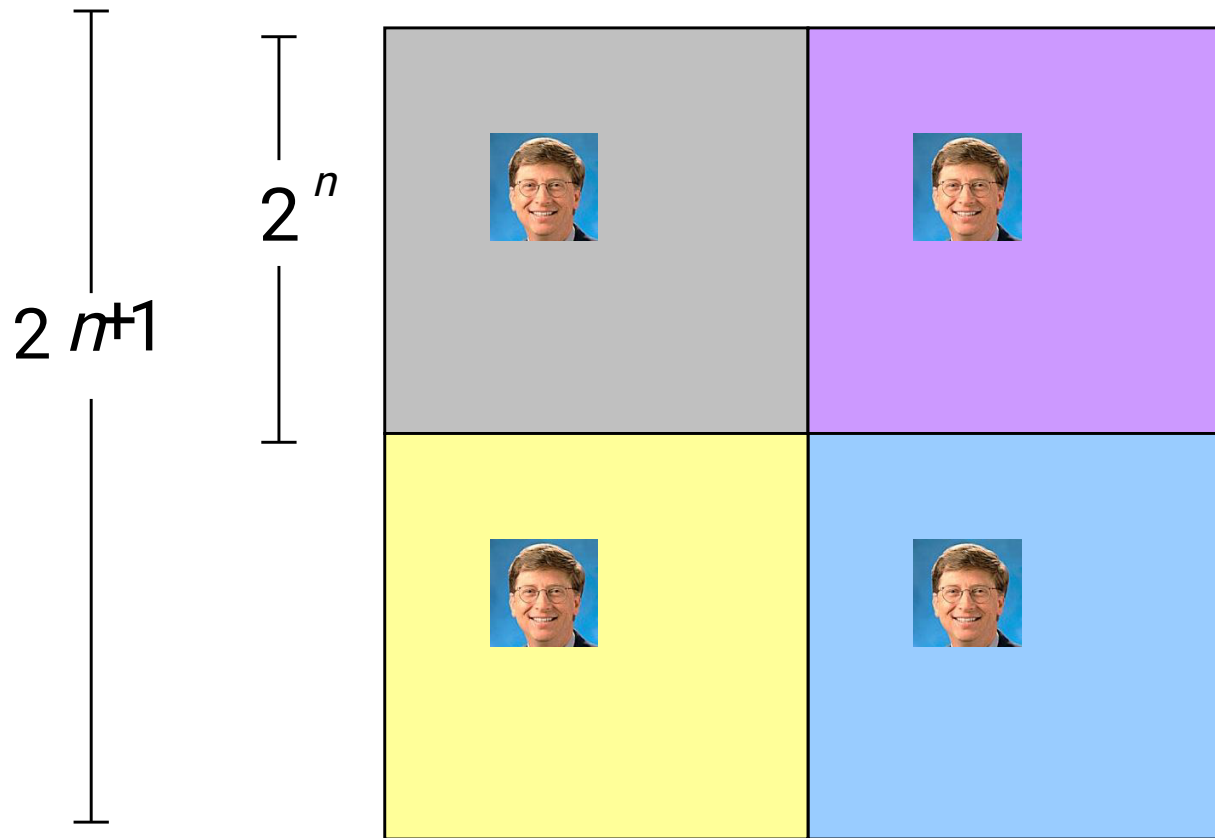
Base case: ($n=0$)



(no tiles needed)

Puzzle

Induction step: assume can tile $2^n \times 2^n$,
prove can handle $2^{n+1} \times 2^{n+1}$.



Puzzle

The new idea:

A stronger property

Prove that we can always find a tiling with Bill
anywhere.

Theorem B: For any $2^n \times 2^n$ puzzle, there is a tiling with Bill
anywhere.

Clearly Theorem B implies Theorem.

Theorem: For any $2^n \times 2^n$ puzzle, there is a tiling with Bill
in the middle.

Puzzle

Theorem B: For any $2^n \times 2^n$ puzzle, there is a tiling with Bill anywhere.

Proof: (by induction on n)

$P(n) ::=$ can tile $2^n \times 2^n$ with Bill anywhere.

Base case: ($n=0$)



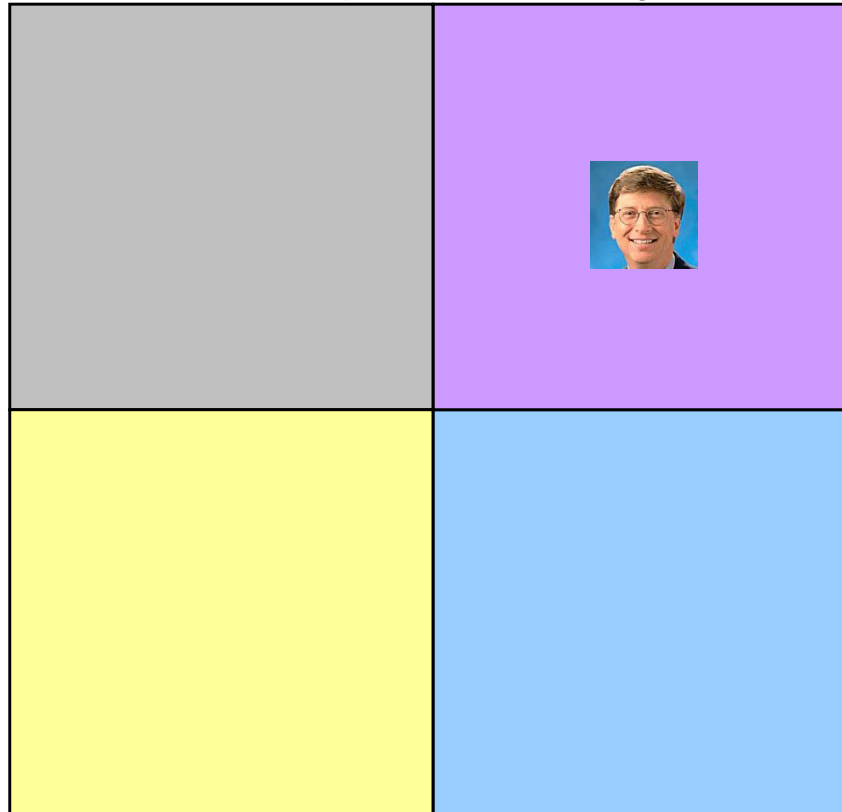
(no tiles needed)

Puzzle

Induction step:

Assume we can get Bill **anywhere in** $2^n \times 2^n$.

Prove we can get Bill anywhere in $2^{n+1} \times 2^{n+1}$.

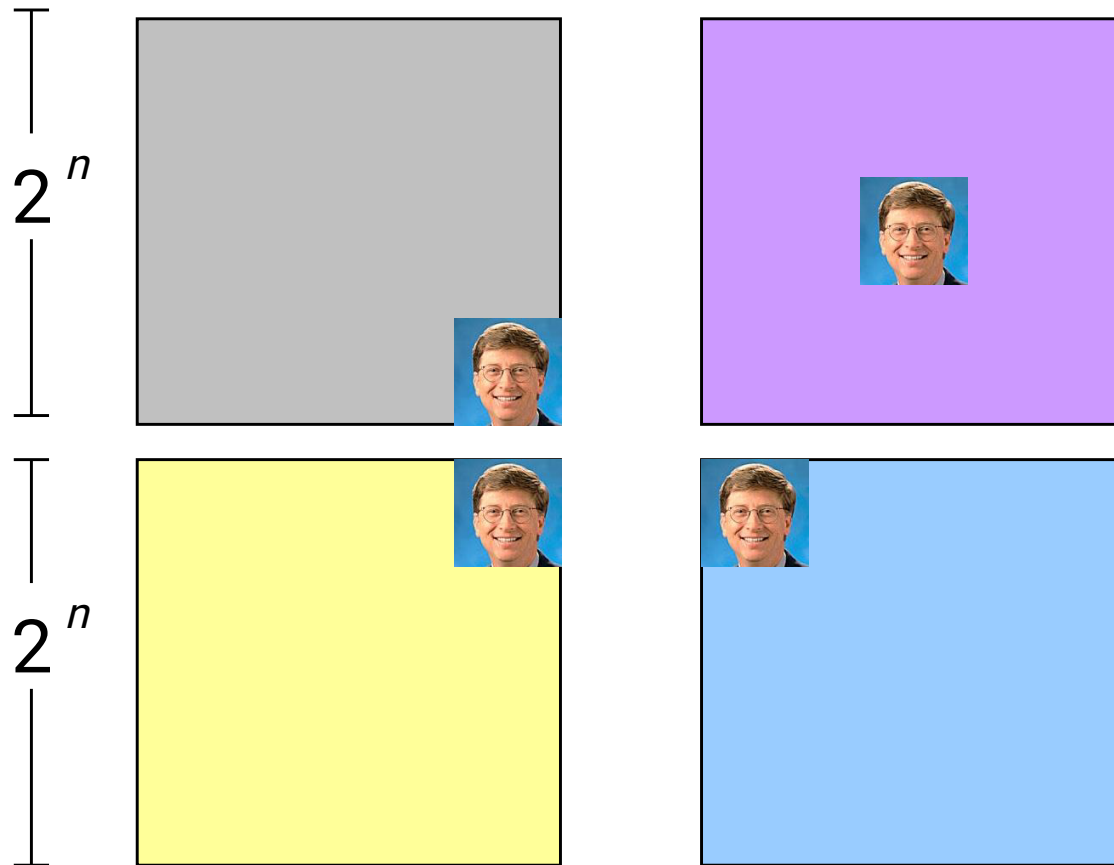


Puzzle

Induction step:

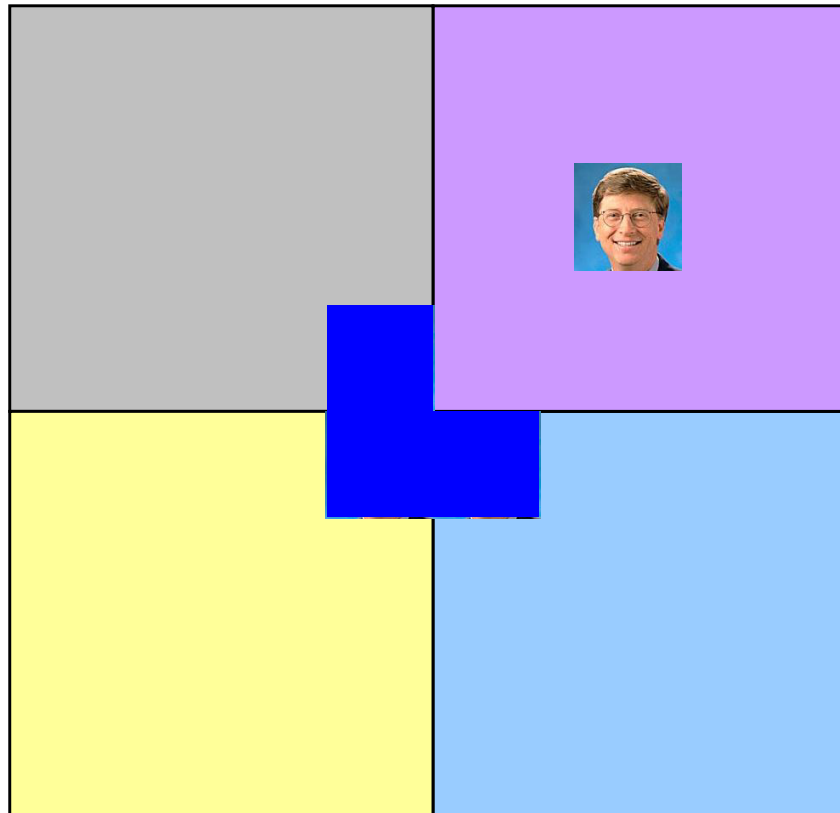
Assume we can get Bill **anywhere in** $2^n \times 2^n$.

Prove we can get Bill anywhere in $2^{n+1} \times 2^{n+1}$.



Puzzle

Method: Now group the squares together,
and fill the center with a tile.



Done!

5.2 Strong induction and well-ordering

- **Strong induction:** To prove $p(n)$ is true for all positive integers n , where $p(n)$ is a propositional function, we complete two steps
- **Basis step:** we verify that the proposition $p(1)$ is true
- **Inductive step:** we show that the conditional statement $(p(1) \wedge p(2) \wedge \dots \wedge p(k)) \rightarrow p(k+1)$ is true for all positive integers k

Strong induction

- Can use all k statements, $p(1)$, $p(2)$, ..., $p(k)$ to prove $p(k+1)$ rather than just $p(k)$
- Mathematical induction and strong induction are equivalent
- Any proof using mathematical induction can also be considered to be a proof by strong induction (induction $\rightarrow$ strong induction)
- It is more awkward to convert a proof by strong induction to one with mathematical induction (strong induction $\rightarrow$ induction)

Strong induction

- Also called the **second principle of mathematical induction** or **complete induction**
- The principle of mathematical induction is called **incomplete induction**, a term that is somewhat misleading as there is nothing incomplete
- Analogy:
 - If we can reach the first step
 - For every integer k , if we can reach all the first k steps, then we can reach the $k+1$ step

Example

- Suppose we can reach the 1st and 2nd rungs 梯级 of an infinite ladder
- We know that if we can reach a rung, then we can reach two rungs higher
- Can we prove that we can reach every rung using the principle of mathematical induction? or strong induction?

Example – mathematical induction

- Basis step: we verify we can reach the 1st rung
- Attempted inductive step: the inductive hypothesis is that we can reach the k -th rung
- To complete the inductive step, we need to show that we can reach $k+1$ -th rung based on the hypothesis
- However, no obvious way to complete this inductive step (because we do not know from the given information that we can reach the $k+1$ -th rung from the k -th rung)

Example – strong induction

- Basis step: we verify we can reach the 1st rung **from the ground**
- Inductive step: the inductive hypothesis states that we can reach each of the first k rungs
- To complete the inductive step, we need to show that we can reach $k+1$ -th rung
- We know that we can reach 2nd rung.
- We note that we can reach the $(k+1)$ -th rung from $(k-1)$ -th rung we can climb 2 rungs from a rung that we already reach
- This completes the inductive step and finishes the proof by strong induction

Show that if n is an integer greater than 1, then n can be written as the product of primes.

Let $P(n)$ be the proposition that n can be written as the product of primes.

- basis: $P(2)$ is true, because 2 can be written as the product of one prime, itself.
- The inductive hypothesis is the assumption that $P(j)$ is true for all integers j with $2 \leq j \leq k$.

- 2 cases: when $k + 1$ is prime and when $k + 1$ is composite.
- If $k + 1$ is prime, we immediately see that $P(k + 1)$ is true. Otherwise, $k + 1$ is composite and can be written as the product of two positive integers a and b with $2 \leq a \leq b < k + 1$. Because both a and b are integers at least 2 and not exceeding k , we can use the inductive hypothesis to write both a and b as the product of primes

Which one to use

- Try to prove with mathematical induction first
- Unless you can clearly see the use of strong induction for proof

5.3 Recursive definitions and structural induction

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A recursively defined picture

Recursive definitions

- The sequence of powers of 2 is given by $a_n = 2^n$ for $n=0, 1, 2, \dots$
- Can also be defined by $a_0=1$, and a rule for finding a term of the sequence from the previous one, i.e., $a_{n+1} = 2a_n$
- Can use induction to prove results about the sequence
- **Structural induction:** We define a set recursively by specifying some initial elements in a basis step and provide a rule for constructing new elements from those already in the recursive step

Recursively defined functions

- Use two steps to define a function with the set of non-negative integers as its domain
- **Basis step:** specify the value for the function at zero
- **Recursive step:** give a rule for finding its value at an integer from its values at smaller integers
- Such a definition is called a recursive or inductive definition

Example

- Suppose f is defined recursively by

- $f(0)=3$

- $f(n+1)=2f(n)+3$

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$

- $f(1)=2f(0)+3=2*3+3=9$

- $f(2)=2f(1)+3=2*9+3=21$

- $f(3)=2f(2)+3=2*21+3=45$

- $f(4)=2f(3)+3=2*45+3=93$

Example

- Give an inductive definition of the factorial function $f(n)=n!$
- Note that $(n+1)!=(n+1) \cdot n!$
- We can define $f(0)=1$ and $f(n+1)=(n+1)f(n)$
- To determine a value, e.g., $f(5)=5!$, we can use the recursive function

$$\begin{aligned} f(5) &= 5 \cdot f(4) = 5 \cdot 4 \cdot f(3) = 5 \cdot 4 \cdot 3 \cdot f(2) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot f(1) \\ &= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot f(0) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = 120 \end{aligned}$$

Recursive functions

- Recursively defined functions are **well defined**
- For every positive integer, the value of the function is determined in an unambiguous way
- Given any positive integer, we can use the two parts of the definition to find the value of the function at that integer
- We obtain the same value no matter how we apply two parts of the definition

Example

- Given a recursive definition of a^n , where a is a non-zero real number and n is a non-negative integer
- Note that $a^{n+1} = a \cdot a^n$ and $a^0 = 1$
- These two equations uniquely define a^n for all non-negative integer n

Example

- Given a recursive definition of $\sum_{k=0}^n a_k$
- The first part of the recursive definition

$$\sum_{k=0}^0 a_k = a_0$$

- The second part is

$$\sum_{k=0}^{n+1} a_k = \left(\sum_{k=0}^n a_k \right) + a_{n+1}$$

Example – Fibonacci numbers

- Fibonacci numbers f_0, f_1, f_2 , are defined by the equations, $f_0=0$, $f_1=1$, and $f_n=f_{n-1}+f_{n-2}$ for $n=2, 3, 4, \dots$

- By definition

$$f_2=f_1+f_0=1+0=1$$

$$f_3=f_2+f_1=1+1=2$$

$$f_4=f_3+f_2=2+1=3$$

$$f_5=f_4+f_3=3+2=5$$

$$f_6=f_5+f_4=5+3=8$$

Example

- Use strong induction to show when $n \geq 3$, $f_n > \phi^{n-2}$ where f_n is a Fibonacci number and $\phi = (1 + \sqrt{5}) / 2$
- Let $p(n)$ be the proposition that $f_n > \phi^{n-2}$
- Basis step: note that

$$\phi < 2 = f_3, \phi^2 = (3 + \sqrt{5}) / 2 < 3 = f_4$$
 so that $p(3)$ and $p(4)$ are true
- Inductive step: assume $p(j)$ is true, i.e., $f_j > \phi^{j-2}$ with $3 \leq j \leq k$ where $k \geq 4$. We need to show that $p(k+1)$ is true, i.e., $f_k > \phi^{k-2}$

Example

- First note that $\diamond \diamond$ is a solution to $x^2 - x - 1 = 0$, so $\diamond \diamond^2 = \diamond \diamond + 1$, thus

$$a^{k-1} = a^2 a^{k-3} = (a + 1) a^{k-3} = a^{k-2} + a^{k-3}$$

- By inductive hypothesis, if $k \geq 4$, it follows

$$f_{k-1} > \diamond \diamond^{k-3}, f_k > \diamond \diamond^{k-2}$$

- So, $f_{k+1} = f_k + f_{k-1} > a^{k-2} + a^{k-3} = a^{k-1}$

It follows that $p(k+1)$ is true. This completes the proof

Recursively defined sets and structures

- Consider the subset S of the set of integers defined by
 - Basis step: $3 \in S$
 - Recursive step: if $x \in S$ and $y \in S$, then $x + y \in S$
- The new elements formed by this are $3+3=6$, $3+6=9$, $6+6=12$, ...
- We will show that S is the set of all positive multiples of 3 (using structural induction)

String

- The set Σ^* of strings over the alphabet Σ can be defined recursively by
 - Basis step: $\epsilon \in \Sigma^*$ (where ϵ is the empty string containing no symbols)
 - Recursive step: if $w \in \Sigma^*$ and $x \in \Sigma$ then $wx \in \Sigma^*$
- The basis step defines that the empty string belongs to string
- The recursive step states new strings are produced by adding a symbol from Σ to the end of strings in Σ^*
- At each application of the recursive step, strings containing one additional symbol are generated

Example

- If $\Sigma = \{0, 1\}$, the strings found to be in Σ^* , the set of all bit strings, are
- $\diamond ? \diamond ?$, specified to be in Σ^* in the basis step
- 0 and 1 found in the 1st recursive step
- 00, 01, 10, and 11 are found in the 2nd recursive step, and so on

Concatenation

- Two strings can be combined via the operation of concatenation
- Let Σ be a set of symbols and Σ^* be the set of strings formed from symbols in Σ
- We can define the concatenation for two strings by recursive steps
 - Basis step: if $w \in \Sigma^*$, then $w \cdot \diamond \diamond = w$, where $\diamond \diamond$ is the empty string
 - Recursive step: If $w_1 \in \Sigma^*$, $w_2 \in \Sigma^*$ and $x \in \Sigma$, then $w_1 \cdot (w_2 x) = (w_1 \cdot w_2)x$
 - Oftentimes $w_1 \cdot w_2$ is rewritten as $w_1 w_2$
 - e.g., $w_1 = \text{abra}$, and $w_2 = \text{cadabra}$, $w_1 w_2 = \text{abracadabra}$

Length of a string

- Give a recursive definition of $L(w)$, the length of a string w
- The length of a string is defined by
 - $L(\diamond \diamond) = 0$
 - $L(wx) = L(w) + 1$ if $w \in \Sigma^*$ and $x \in \Sigma$

Well-formed formulae

- We can define the set of **well-formed formulae** for compound statement forms involving T, F, proposition variables and operators from the set $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$
- Basis step: T, F, and s, where s is a propositional variable are well-formed formulae
- Recursive step: If E and F are well-formed formulae, then $\neg E$, $E \wedge F$, $E \vee F$, $E \rightarrow F$, $E \leftrightarrow F$ are well-formed formulae
- From an initial application of the recursive step, we know that $(p \vee q)$, $(p \rightarrow F)$, $(F \rightarrow q)$ and $(q \wedge F)$ are well-formed formulae
- A second application of the recursive step shows that
- $((p \vee q) \rightarrow (q \wedge F))$, $(q \vee (p \vee q))$, and $((p \rightarrow F) \rightarrow T)$ are well-formed formulae

Well-Formed Formulae of Operators and Operands

- BASIS STEP: x is a well-formed formula if x is a numeral or a variable.
- RECURSIVE STEP: If F and G are well-formed formulae, then $(F + G)$, $(F - G)$, $(F * G)$,
- (F/G) , and $(F \uparrow G)$ are well-formed formulae.

Rooted trees

- The set of rooted trees, where a rooted tree consists of a set of vertices containing a distinguished vertex called the root, and edges connecting these vertices, can be defined recursively by
 - Basis step: a single vertex r is a rooted tree
 - Recursive step: suppose that $T_1, T_2, \dots, T_n$ are disjoint rooted trees with roots $r_1, r_2, \dots, r_n$, respectively.
 - Then the graph formed by starting with a root r , which is not in any of the rooted trees $T_1, T_2, \dots, T_n$, and adding an edge from r to each of the vertices $r_1, r_2, \dots, r_n$, is also a rooted tree

Rooted trees

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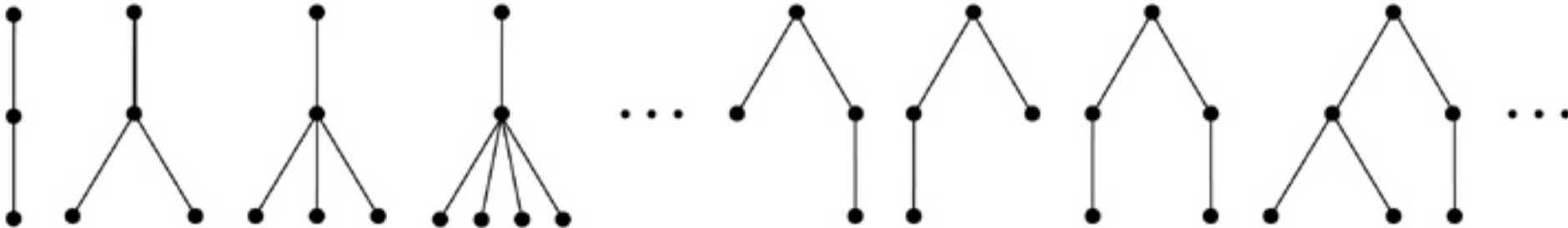
Basis step



Step 1



Step 2



Binary trees

- At each vertex, there are at most two branches (one left subtree and one right subtree)
- Extended binary trees: the left subtree or the right subtree can be empty
- Full binary trees: must have left and right subtrees

Extended binary trees

- The set of extended binary trees can be defined by
 - Basis step: the empty set is an extended binary tree
 - Recursive step: If T_1 and T_2 are disjoint extended binary trees, there is an extended binary tree, denoted by $T_1 \cdot T_2$, consisting of a root r together with edges connecting the root to each of the roots of the left subtree T_1 and right subtree T_2 , when these trees are non-empty

Extended binary trees

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Basis step

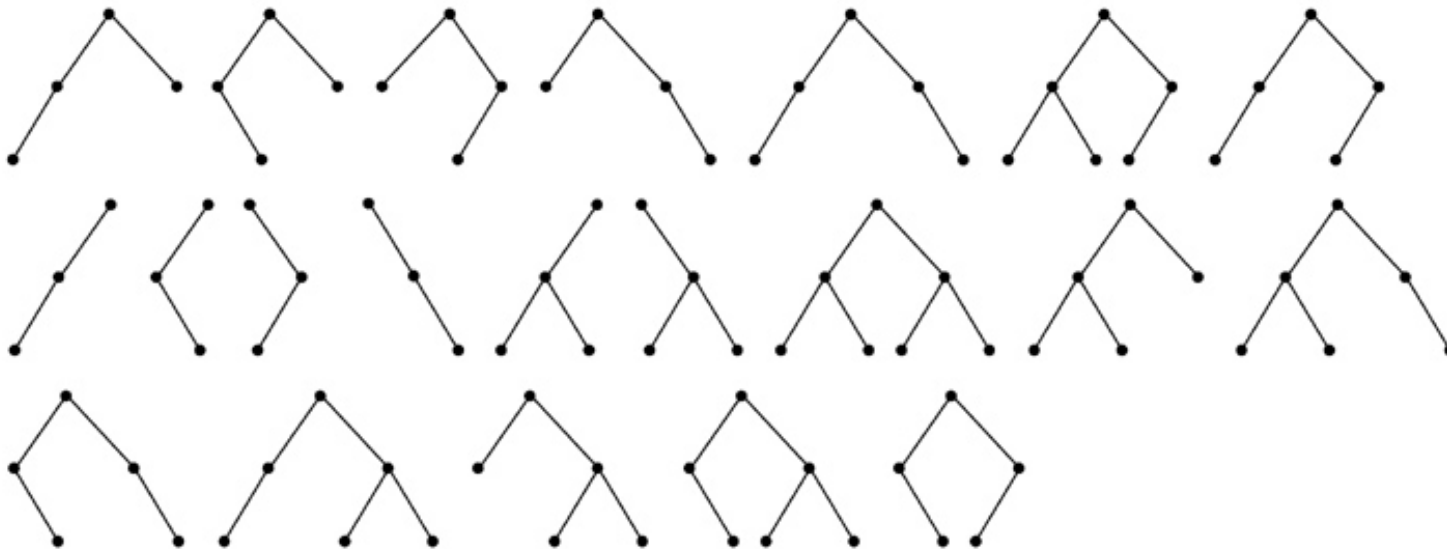
Step 1



Step 2



Step 3



Full binary trees

- The set of full binary trees can be defined recursively
 - Basis step: There is a full binary tree consisting only of a single vertex r
 - Recursive step: If T_1 and T_2 are disjoint full binary trees, there is a full binary tree, denoted by $T_1 \cdot T_2$, consisting of a root r together with edges connecting the root to each of the roots of the left subtree T_1 and right subtree T_2

Full binary tree

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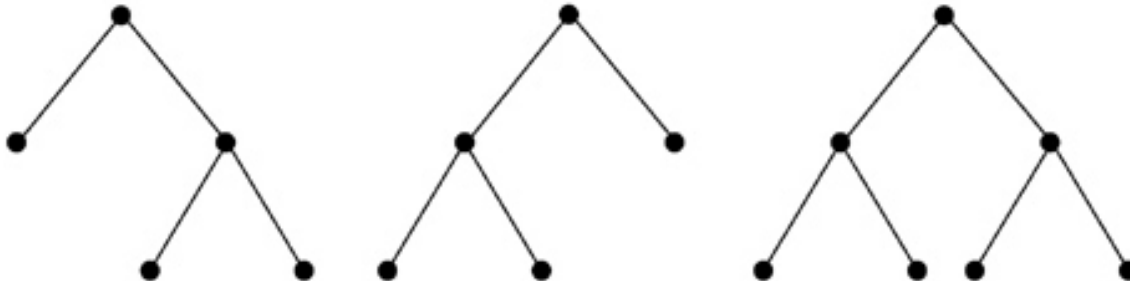
Basis step



Step 1



Step 2



structural induction

- BASIS STEP: Show that the result **holds for all elements** specified in the **basis step** of the recursive definition to be in the set.
- RECURSIVE STEP: Show that if the statement is true for **each of** the elements used to construct new elements in the **recursive step** of the definition, the result holds for these new elements.

Structural induction

- Show that the set S defined by
 - $3 \in S$ and
 - if $x \in S$ and $y \in S$, then $x+y \in S$,is the set of multiples of 3
- Let A be the set of all positive integers divisible by 3
- To prove $A=S$, we must show that $A \subseteq S$, and $S \subseteq A$
- To show $A \subseteq S$, we must show that every positive integer divisible by 3 is in S
- Use mathematical induction to prove it

Structural induction

- Let $p(n)$ be the statement that $3n$ belongs to S
- Basis step: it holds as the first part of recursive definition of S , $3 \cdot 1 = 3 \in S$
- Inductive step: assume that $p(k)$ is true, i.e., $3k$ is in S . As $3k \in S$ and $3 \in S$, it follows from the 2nd part of the recursive definition of S that $3k+3=3(k+1) \in S$. So $p(k+1)$ is true

Structural induction

- To show that $S \subseteq A$, we use recursive definition of S
- The basis step of the definition specifies that 3 is in S
- As $3=3 \cdot 1$, all elements specified to be in S in this step are divisible by 3, and there in A
- To finish the proof, we need to show that all integers in S generated using the 2nd part of the recursive definition are in A
- This consists of showing that $x+y$ is in A whenever x and y are elements of S also assumed to be in A
- If x and y are both in A , it follows that $3|x$, $3|y$, and thus $3|x+y$, thereby completing the proof

Trees and structural induction

- To prove properties of trees with structural induction
 - Basis step: show that the result is true for the tree consisting of a single vertex
 - Recursive step: show that if the result is true for the trees T_1 and T_2 , then it is true for $T_1 \cdot T_2$, consisting of a root r , which has T_1 as its left subtree and T_2 as its right subtree

Height of binary tree

- We define the height $h(T)$ of a full binary tree T recursively
 - Basis step: the height of the full binary tree T consisting of only a root r is $h(T)=0$
 - Recursive step: If T_1 and T_2 are full binary trees, then the full binary tree $T = T_1 \cdot T_2$ has height $h(T)=1+\max(h(T_1), h(T_2))$

Number of vertices in a binary tree

- If we let $n(T)$ denote the number of vertices in a full binary tree, we observe that $n(T)$ satisfies the following recursive formula:
 - Basis step: the number of vertices $n(T)$ of the full binary tree consisting of only a root r is $n(T)=1$
 - Recursive step: If T_1 and T_2 are full binary trees, then the number of vertices of the full binary tree $T = T_1 \cdot T_2$ is $n(T)=1+n(T_1)+n(T_2)$

Theorem

- If T is a full binary tree T , then $n(T) \leq 2^{h(T)+1} - 1$
- Use structural induction to prove this
- Basis step: for the full binary tree consisting of just the root r the result is true as $n(T)=1$ and $h(T)=0$, so $n(T)=1 \leq 2^{0+1} - 1 = 1$

$$n(T_1) \leq 2^{h(T_1)+1} - 1, \quad n(T_2) \leq 2^{h(T_2)+1} - 1$$

- Inductive step: For the inductive hypothesis we assume that
where T_1 and T_2 are full binary trees

Theorem

- By the recursive formulae for $n(T)$ and $h(T)$, we have $n(T)=1+n(T_1)+n(T_2)$ and $h(T)=1+\max(h(T_1), h(T_2))$

- Thus,

$$\begin{aligned}n(T) &= 1 + n(T_1) + n(T_2) \\&\leq 1 + (2^{h(T_1)+1} - 1) + (2^{h(T_2)+1} - 1) \\&\leq 2 \cdot \max(2^{h(T_1)+1}, 2^{h(T_2)+1}) - 1 \\&= 2 \cdot 2^{\max(h(T_1), h(T_2))+1} - 1 \\&= 2 \cdot 2^{h(T)} - 1 \\&= 2^{h(T)+1} - 1\end{aligned}$$

- This completes the inductive step

exercise

- Determine which Fibonacci numbers are divisible by 3. Use a form of mathematical induction to prove your conjecture.
- Let S be the subset of the set of ordered pairs of integers defined recursively by

Basis step: $(0, 0) \in S$.

Recursive step: If $(a, b) \in S$, then $(a + 2, b + 3) \in S$ and $(a + 3, b + 2) \in S$.

Use strong induction and structural induction to show $5 \mid a + b$ when $(a, b) \in S$.